MHF 4U1

CHAPTER 2: POLYNOMIAL EQUATIONS AND INEQUALITIES

Day	Section	Homework
2	2.1 Division of Polynomials & The Remainder Theorem	Day 1: p. 91: #1-4 eo,5,6,16 Day 2: p. 91: #9eo,11,13,15,21, 22
2	2.2 The Factor Theorem	Day 1: p. 102 #2,3,5, 8-10,18, 19 Day 2: p. 102 #4,12,13,15,20
1	2.3 Polynomial Equations	Page 110 #(2-4)eo, 5 (c is true), (7-8)eo,18
1	2.4 Families of Polynomial Functions	Page 119 #1-3, 7, 8, 10-12, 14, 16
1	2.3/2.4 Applications of Zeros	Page 110 # 10, 11, 14, 17, 19, 21 Page 119 #18ab, 20 Page 146 #8
1	2.6 Solving Inequalities Algebraically	Page 129 #1, 5 Page 138 # 1, 4, 5, 6bc, 7, 9, 14
2	Review Practice Test	p. 140 # 1 - 9, 10a, 12 - 14, 15bd, 17, 18a p. 142 # 1 - 8, 9ab, 11, 12, 16, 17
1	CHAPTER 2 TEST	

CHAPTER 2 TEST DATE:

Recall: long division $6\overline{)493}$

Division Statement: 493 =

This method also applies when dividing polynomials.

Ex 1: Divide $x^3 + 4x^2 - 3x + 4$ by $x - 1$. State restrictions and write the division statement.

Ex 2: Divide, state restrictions and write the division statement:

$$x+2\overline{)2x^4-3x^2+2}$$

Ex 3: Divide $x^4 + 5x - 8$ by $x - 2$. Give the Division statement & state restrictions.

Ex 4: Divide $3x^3 - 11x^2 + 21x - 8$ by $3x - 2$.

Ex 5: Divide $x^4 + x^3 - x^2 + 7$ by $x^2 + 3x - 1$.

SHORTCUT: Synthetic Division;

Redo Ex 1: Divide $x^3 + 4x^2 - 3x + 4$ by $x - 1$

division statement: $x^3 + 4x^2 - 3x + 4 =$

Redo Ex 2: Divide $x + 2\overline{)2x^4 - 3x^2 + 2}$

Redo Ex 3: Divide $x^4 + 5x - 8$ by $x - 2$

Redo Ex 5:

* **Redo Ex 4**: Divide $3x^3 - 11x^2 + 21x - 8$ by $3x - 2$.

$$\frac{2}{3} \;\Big|\; 3 \quad -11 \quad 21 \quad -8$$

$(3x - 2) = 0$

$3(x - \frac{2}{3}) = 0$

we still need to factor out the **3**!

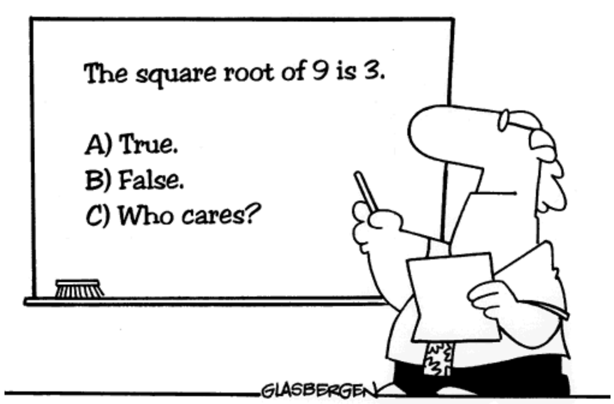
Therefore, $3x^3 - 11x^2 + 21x - 8$

$= (3x - 2)\left(\dfrac{3x^2 - 9x + 15}{\mathbf{3}}\right) + 2$

$= (3x - 2)(\mathbf{x^2 - 3x + 5}) + 2$

Try it now!

When $12x^3-14x^2+px-1$ is divided by $qx-2$, the quotient is $4x^2+rx+3$ and the remainder is 5. Find p, q, and r.



Many students actually look forward to Mr. Atwadder's math tests.

Ex 1: Find the remainder when $(x^4 + 5x - 8) \div (x - 2)$.

Ex 2: Given $f(x) = x^4 + 5x - 8$, find $f(2)$.

THE REMAINDER THEOREM:
When the function $f(x)$ is divided by $(x - \mathbf{p})$, the remainder is $f(\mathbf{p})$.

Ex 3: Find the remainder when $4x^2 + 5x - 1$ is divided by $2x - 1$.

Ex 4: When $x^3 + \mathbf{k}x^2 + 7x + 5$ is divided by $x - 3$, the remainder is 8. Find **k**.

Ex 5: When $\mathbf{a}x^3 - 3x^2 + \mathbf{b}x + 2$ is divided by $x + 3$, the remainder is 11. When it is divided by $x - 2$, the remainder is -4. Find **a** and **b** algebraically.

2.2 THE FACTOR THEOREM

Q: Is 5 a factor of 90?

A: YES, because it divides evenly into 90. (ie. the remainder is ZERO!)

Q: When will $(x - b)$ be a factor of a polynomial function, $f(x)$?

A: When it divides evenly into $f(x)$, that is, when the remainder is ZERO!

The FACTOR THEOREM:

A polynomial function, $f(x)$, has $(x - \mathbf{b})$ as a factor if and only if $f(\mathbf{b}) = \mathbf{0}$.

Ex 1: Show that $x + 3$ is a factor of $4x^3 + 16x^2 + 9x - 9$.

Ex 2: Is $t - 5$ a factor of $2t^3 - t^2 - 3t + 1$?

RATIONAL ZERO THEOREM:

If $(\mathbf{a}x - \mathbf{b})$ is a factor of the polynomial function $f(x)$, then $f\left(\frac{b}{a}\right) = 0$.

Furthermore, **a** must be a factor of the leading coefficient of $f(x)$ and **b** must be a factor of the constant term of $f(x)$.

Ex 3: Factor $x^3 - 6x^2 + 11x - 6$.

Ex 4: Factor $3x^3 + 8x^2 + 3x - 2$.

2.2 THE FACTOR THEOREM: DAY 2

Ex 1: Use the factor theorem to factor $m^3 - 8$.

b	a

*In fact, this can be generalized to factor all functions of the form $\mathbf{x^3 - y^3}$.

Recall: **Difference of Squares:** $\mathbf{x^2 - y^2 = (x + y)(x - y)}$

There is NO Sum of Squares ! $\mathbf{x^2 + y^2}$ = not factorable !

> ***New**: **Difference of Cubes:**
> $\mathbf{x^3 - y^3 = (x - y)(x^2 + xy + y^2)}$
>
> **Sum of Cubes:**
> $\mathbf{x^3 + y^3 = (x + y)(x^2 - xy + y^2)}$

Ex 2: Factor $27x^3 + 64y^3 =$

Ex 3: Use the factor Theorem to factor $4x^3 + 8x^2 - 9x - 18$.

Note: There are actually ______ possible values of $\frac{b}{a}$. **Sometimes**, we can avoid the tedious process of finding $f(\frac{b}{a}) = 0$. Instead, we can **FACTOR BY GROUPING** terms:

$\mathbf{4x^3 + 8x^2 - 9x - 18}$

$= \mathbf{4x^2}(\mathbf{x + 2}) - \mathbf{9}(\mathbf{x + 2})$

$= (\mathbf{x + 2})(\mathbf{4x^2 - 9})$

$= (x + 2)(2x + 3)(2x - 3)$

Ex 4: Factor using the most appropriate method.

a) $m^3 + 3m^2 - 25m - 75$

b) $2kp^3 - p^3 - 16k + 8$

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Try it now!

Find the value of a and b if x^2-5x+4 is a factor of the polynomial $2x^3+ax^2+bx-4$. Express the polynomial in factored form.

3		1				9		7
	4			6			2	
5		2				8		3
			1		4			
	7			2			3	
			7		6			
1		7				4		8
	2			5			7	
6		5				3		2

2.2 FACTORING REVIEW

Factoring: - writing a polynomial as a product

- the *opposite* of expanding

Strategies for factoring fully:

- Look for a greatest common factor **(GCF)**, then divide each term by it. **DO THIS FIRST!!!**
- For trinomials ($ax^2 + bx + c$) use: product and sum, "criss-cross", etc.
- For difference of squares use: $\mathbf{a^2 - b^2 = (a + b)(a - b)}$
- For perfect squares use: $\mathbf{a^2 + 2ab + b^2 = (a + b)^2}$ OR $\mathbf{a^2 - 2ab + b^2 = (a - b)^2}$
- For difference of cubes use: $\mathbf{x^3 - y^3 = (x - y)(x^2 + xy + y^2)}$
- For sum of cubes use: $\mathbf{x^3 + y^3 = (x + y)(x^2 - xy + y^2)}$
- For polynomials with more than 3 terms: first try **grouping** *(may need to rearrange),* then use the **Factor Theorem** if necessary.

Ex 1: Factor.

a) $12x^3y^4 - 4x^2y^2 + 8x^3y^5$

b) $2a(5b - 1) - 4(5b - 1)$

c) $2x^2 - 16x + 30$

d) $8y^2 - 10y + 3$

e) $4x^2 - 12x + 9$

f) $16x - 9x^3$

g) $8s^3 + 125t^3$

h) $5p^6 - 5$

Ex 3: Factor.

a) $3p^2 + 11pq - 4q^2$

b) $4y^2 - 9(x + 3)^2$

c) $4m(n - 1) + 3(1 - n)$

Ex 4: Factor by grouping.

a) $ax + cx + ay + cy$

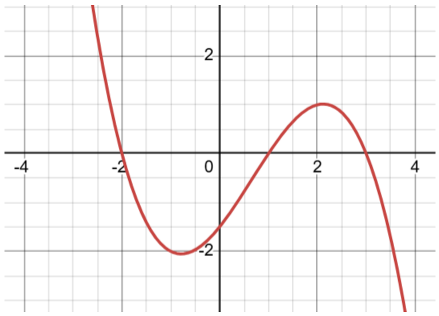
b) $y^3 + y^2 - 9y - 9$

c) $q^2 – p^2 + 2p - 1$

2.3 POLYNOMIAL EQUATIONS

The zeros, or <u>REAL ROOTS</u> of a polynomial equation correspond to the x-intercepts of its graph.

<u>Ex 1</u>: Solve:

a) 

b) $(x + 1)(x - 3)(x^2 + 1) = 0$

c) $x^2 + 5x = 2$

d) $8x^4 - 64x = 0$

e) $x^3 - 3x^2 + x + 5 = 0$

f) $3(x^2 + 2x)^2 + 2(x^2 + 2x) = 1$

2.3/2.4 APPLICATIONS OF ZEROS

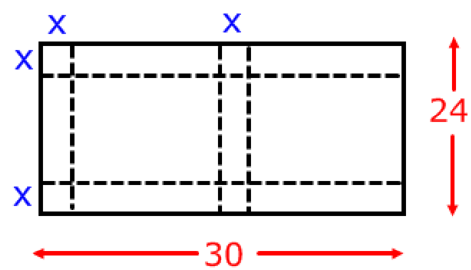
Example 1: Use a factoring strategy to solve for x.

a) $9(x + 1)^2 - 16(2x - 3)^2 = 0$

b) $3(x^2 + 2x)^2 + 2(x^2 + 2x) = 1$

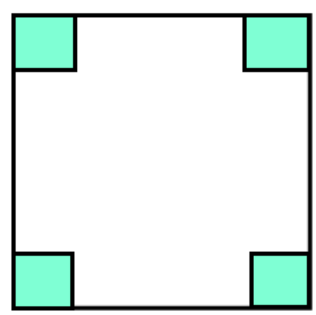
Ex 2: Cardboard sheets that measure **24 cm X 30 cm** are used to make boxes by folding as shown.

a) Express the volume as a function of x.

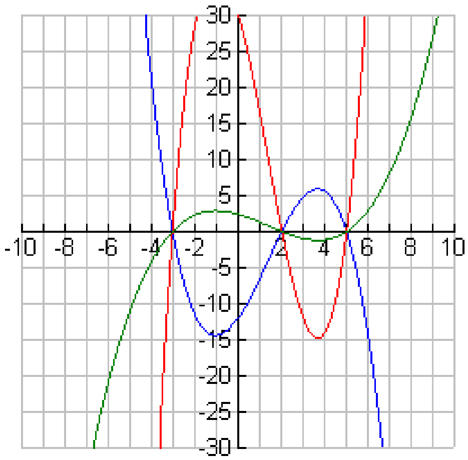


b) Determine the dimensions of the box if the volume is 200 cm^3 (round to one decimal place).

Ex 3: Open top boxes are formed by cutting equal squares from the corners of cardboard sheets measuring **16 cm X 14 cm**. Determine all possible dimensions of the boxes if each has a volume of **240 cm^3**.



2.4 FAMILIES OF POLYNOMIAL EQUAITONS



What similarities do you notice among all these functions?

A **family of functions** is a set of functions that have the same x-intercepts.

Ex 1: a) Determine the equation for the **family of cubics** with zeros: ±1 and 5.

b) Find the cubic that passes through (-3,32).

Ex 2: a) Determine the equation in SIMPLIFIED FORM for the family of quartic functions with zeros $-2 \pm \sqrt{3}$ and $3 \pm \sqrt{5}$.

b) Find the quartic that passes through (-1,-242).

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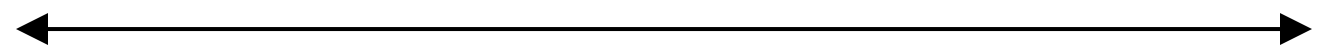
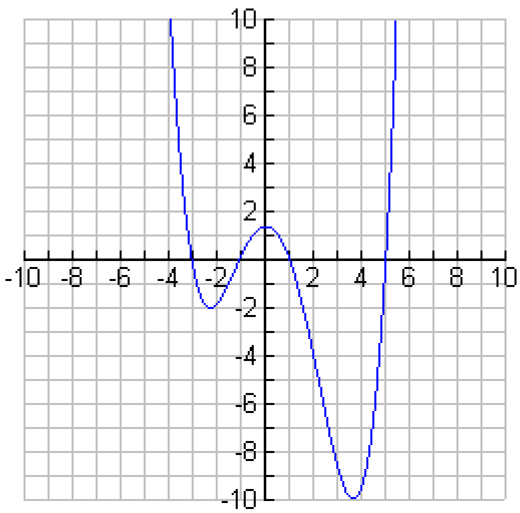
Try it now!

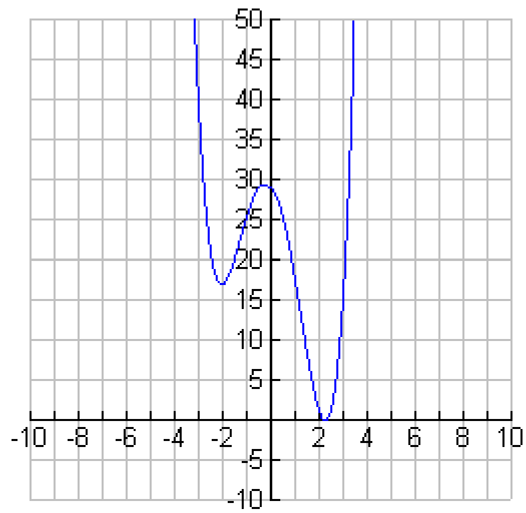
1. **Show on the number line the solution to :**

   3 - 4x < x – 5.



2. **Given the following graphs of functions, determine where f(x) $\leq$ 0.**





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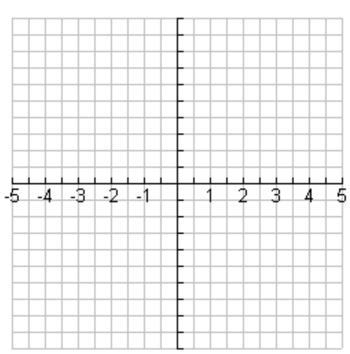
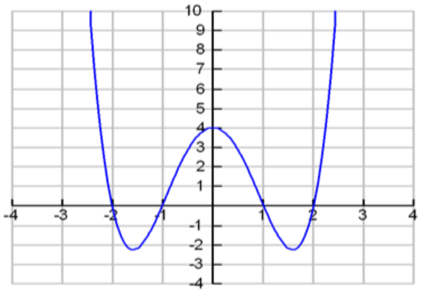
3**. Determine an expression for f(x) in which:**

a) f(x) is a quartic function,
b) f(x) >0 when -2<x<1,
c) f(x) $\leq$ 0 when x<-2 or x>1,
d) f(x) has an order two zero at x=3 **and**
e) f(-1) = 96.

Hint…a sketch may help!



Example:
Solve the inequality $x^4 - 5x^2 + 4 < 0$.



Solution:
$x^4 - 5x^2 + 4 < 0$ is true when the graph is *below the x-axis*. Notice that the 4 intercepts divide the graph into 5 intervals. The intervals that satisfy the inequality are:

$\boxed{-2 < x < -1}$ and $\boxed{1 < x < 2}$

Ex 1: SOLVE the following inequalities.

a) $x + 5 \leq -8$

b) $(x + 4)(x - 1) > 0$

c) $(x - 3)(2x + 1)(x - 5) \geq 0$

d) $-2x^3 + 50x > 0$

e) $5x^3 - 12x^2 \leq 11x - 6$

f) $(x - 1)(x + 2)^2(x - 5) > 0$

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Try it now!

1. If $2x^3 - 9x^2 + 4x - 7$ is divided by $x - 3$ to give a quotient of $2x^2 - 3x - 5$ and a remainder of -22 , then which of the following is true?

A $2x^3 - 9x^2 + 4x - 7 = (x - 3)(2x^2 - 3x - 5) + 22$
B $2x^3 - 9x^2 + 4x - 7 = (x - 3)(2x^2 - 3x - 5) - 22$
C $(x - 3)(2x^2 - 3x - 5) = 22$
D $(x - 3)(2x^2 - 3x - 5) = -22$

2. If $P(x) = 4x^3 - 11x^2 - 6x + 9$ is divided by $x - 3$, what is the remainder?

A $P(3)$
B $P(-3)$
C 0
D A and C

3. For a polynomial $P(x)$, if $P\left(-\frac{3}{5}\right) = 0$, then which of the following must be a factor of $P(x)$?

A $x - \frac{3}{5}$
B $3x + 5$
C $5x + 3$
D $5x - 3$

4. Determine the value of k so that $x - 3$ is a factor of $x^3 - 3x^2 + x + k$.

A $k = 3$
B $k = -3$
C $k = 1$
D $k = -1$

5. Which of the following is the factored form of $x^4 - 2x^2 - 3$?

A $(x - 1)(x + 1)(x - 3)$
B $(x^2 - 1)(x + 3)$
C $(x^2 + 1)(x^2 - 3)$
D none of the above

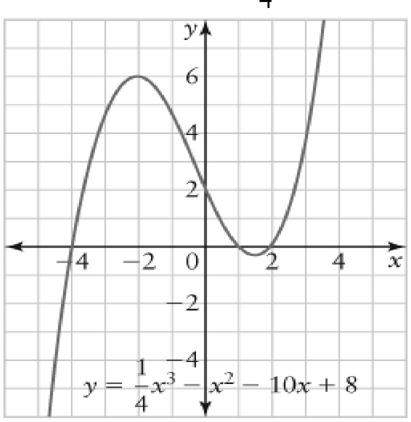
6. What is the maximum number of real distinct roots that a quartic equation can have?

A infinitely many
B 4
C 2
D none of the above

7. How many cubic functions have zeros –10, –5, and 4?

A 1
B 2
C 3
D infinitely many

8. A graph of $y = \frac{1}{4}x^3 + x^2 - 10x + 8$ is shown. Use the graph to solve $x^3 + 4x^2 - 40x + 32 < 0$.



A $x < -4$
B $x < -4, 1 < x < 2$
C $-4 < x < 1, x > 3$
D $x < -4, 1 < x > 2$